VIVEK KUMAR

Backend Software Engineer • Java & Spring Boot • Microservices • AWS Cloud • GenAI / LangChain

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PROFESSIONAL SUMMARY

Backend Software Engineer with hands-on expertise building resilient, distributed microservices, event-driven architectures, and cloud-native backends using Java 17, Spring Boot 3, Apache Kafka, and AWS. Experienced in payment gateway integration (Trustly HMAC-SHA256), Redis caching, and incorporating Generative AI reasoning workflows (Spring AI, LangChain, semantic vector search, and RAG) into enterprise backends. Dual AWS Certified (Developer – Associate & Data Engineer – Associate).

TECHNICAL SKILLS

Core Java & Backend: Java (8/11/17), Spring Boot 3, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, Microservices, MVC

Cloud & DevOps: AWS (ECS Fargate, Lambda, S3, RDS PostgreSQL, CloudWatch), Terraform (IaC), Docker, GitHub Actions CI/CD, Linux

Generative AI & LangChain: LangChain, Spring AI, LangGraph, LCEL, OpenAI APIs, Vector Databases (pgvector, ChromaDB), Semantic RAG Pipelines, Text-to-SQL, Redis Semantic Caching

Messaging & Caching: Apache Kafka, ActiveMQ, Redis (Caching, Geospatial, Pub/Sub), RabbitMQ

Databases & Testing: PostgreSQL, MySQL, MongoDB, Git, Maven, Gradle, JUnit 5, Mockito, Postman, Jira, Agile Scrum

PROFESSIONAL EXPERIENCE

Java Developer Intern | HulkHire (Remote) Jul 2024 – Dec 2024

- Engineered enterprise backend microservices in Java 17 & Spring Boot with global exception handling, validation frameworks, and RESTful APIs.
- Integrated Trustly Payment Gateway workflows featuring HMAC-SHA256 signature verification, duplicate transaction filters, and idempotency.
- Implemented Redis caching strategies to offload frequent database queries, slashing database load and decreasing latency by 40%.
- Configured Kafka topics and ActiveMQ message brokers for reliable asynchronous event delivery across payment and notification services.
- Developed an automated SQL query explanation and performance optimization bot using LangChain and OpenAI LLMs to assist developers in reviewing complex Hibernate queries.
- Authored robust unit and integration test suites using JUnit 5 and Mockito, ensuring over 85% code test coverage.

KEY PROJECTS

Enterprise GenAI Knowledge Assistant & RAG Microservice | Java 17, Spring Boot, Spring AI, Python, LangChain, pgvector, PostgreSQL, Docker GitHub

- Architected enterprise knowledge extraction microservice integrating Spring Boot with a Python LangChain LCEL backend for semantic document Q&A; and policy retrieval.
- Implemented semantic vector search using pgvector in PostgreSQL with hybrid BM25 search, dynamic prompt templates, and conversational context retention.
- Integrated Redis semantic caching for LLM responses, eliminating duplicate external model calls and slashing LLM API operating costs by 65%.

Distributed Ride-Hailing Backend Microservice (Oxima) | Java 17, Spring Boot 3, Apache Kafka, Redis Geospatial, WebSockets, PostgreSQL, Docker GitHub

- Architected high-throughput backend services managing driver dispatching, fare calculation, and real-time state synchronization.
- Utilized Redis Geospatial indexes (GEOADD, GEORADIUS) to achieve sub-50ms driver matching and Kafka topic partitions for guaranteed ordered event processing.
- Secured microservice communications with Spring Security, JWT role-based access control, and WebSocket STOMP handshake authentication.

AWS ECS Fargate TicketDesk Cloud Infrastructure | AWS ECS Fargate, Terraform, Docker, AWS RDS PostgreSQL, GitHub Actions, FastAPI GitHub

- Provisioned production-grade cloud infrastructure using Terraform (IaC): custom VPC, multi-AZ subnets, ALB, and auto-scaling ECS Fargate containers.
- Automated continuous integration and zero-downtime rolling deployments using GitHub Actions with AWS OIDC authentication and Amazon ECR.
- Integrated AWS CloudWatch dashboards and health check alarms for proactive uptime and error rate monitoring.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science & Engineering | DIT University, Dehradun 2021 – 2025

Certifications: AWS Certified Developer – Associate (DVA-C02) • AWS Certified Data Engineer – Associate (DEA-C01) • Node.js & Express (Coursera) • React Frontend Development (freeCodeCamp) • Agile Project Management (LinkedIn Learning)